**UNIVERSITY OF TIRANA**
**FACULTY OF ECONOMICS**
**DEPARTMENT OF STATISTICS AND APPLIED INFORMATICS**

# Data Security and Behavior of Minors on Social Media Platforms in Albania:

# The Case of TikTok

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First Cycle Of Studies

Diploma: Bachelor in Economic Informatics

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**Tirana,** 2025

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Kamelisa Selami
June ,2025

# Abstract

***Data Security and Behavior of Minors on Social Media Platforms in Albania: The Case of TikTok***

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This thesis explores the intersection between data privacy and adolescent digital behavior in the context of Albania's 2024 national ban on TikTok. With minors representing a significant portion of social media users, the study investigates how Albanian teenagers navigate data collection practices, react to platform restrictions, and engage with privacy protections. Drawing from a cross-sectional survey of 203 students in Tirana and supported by policy analysis, the research finds that 84.5% of minors continued using TikTok post-ban, mainly through virtual private networks, demonstrating the limitations of access-based enforcement.

The findings reveal a mixed landscape: while some teenagers exhibit protective behaviors, such as using fake email addresses, emotional responses vary widely, and harmful content remains underreported. Moreover, a clear "knowledge-action" gap persists, where awareness of data practices does not consistently translate into safer behavior. The study evaluates Albania's evolving legal framework, including recent cybersecurity and data protection legislation, and contrasts it with broader European regulatory trends. It concludes that bans alone are insufficient without integrated education, transparent governance, and youth-centered digital literacy initiatives. This thesis contributes new empirical insights and offers actionable policy recommendations tailored to the Albanian context.

# Acknowledgments

This thesis would not have been possible without the support, encouragement, and belief of many people who stood by me during this challenging and meaningful journey.

First and foremost, I want to express my deepest gratitude to my academic supervisor, Dr.Matilda Veliu for guiding me with patience and precision. Your feedback, clarity, and belief in the importance of this topic motivated me to keep pushing forward, even when the process felt overwhelming.

A special thank you goes to the directors, teachers, and students of Myslym Keta High School and Betim Muço Elementary School, who opened their doors to this research and contributed with honesty and sincerity. Your participation gave real meaning to the data behind this thesis, and I am sincerely grateful for your cooperation and trust.

To the professors and mentors, those who did not just teach, but believed in me, even when I sometimes doubted myself, thank you for seeing potential in me and pushing me to own it.

To my sister, who has been my personal cheerleader and safe space, thank you for your unconditional support, and for reminding me that I am capable, even when I forget. This thesis carries a piece of you in every paragraph I wrote with more strength than I thought I had.

Lastly, to my family and friends, thank you for standing by me during sleepless nights, stressful deadlines, and moments of self-doubt. Your support grounded me. This work is not just a personal milestone, but a reflection of the people who walked beside me every step of the way.

**Faleminderit, from the heart**.

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# List of Abbreviations

| Abbreviation | Full Term |
|---|---|
| VPN | Virtual Private Network |
| GDPR | General Data Protection Regulation |
| CSIRT | Computer Security Incident Response Team |
| DPIA | Data Protection Impact Assessment |
| OECD | Organisation for Economic Co-operation and Development |
| EU | European Union |
| SNI | Server Name Indication |
| DPC | Data Protection Commission (Ireland) |
| ICO | Information Commissioner’s Office (UK) |
| MIL | Media and Information Literacy |
| UI | User Interface |
| AI | Artificial Intelligence |
| ISO | International Organization for Standardization |
| NIST | National Institute of Standards and Technology |
| SCiDEV | Centre Science and Innovation for Development |
| T&C | Terms and Conditions |
| DSA | Digital Services Act |
| UNESCO | United Nations Educational, Scientific and Cultural Organization |

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# I. INTRODUCTION

## I.2 Setting the stage

Over the past decade, social media platforms have reshaped how teenagers express themselves, build community, and find information. In Albania, where more than eighty percent of households have Internet access, TikTok became one of the fastest-growing mobile applications. By early 2024, TikTok's parent company ByteDance reported that one point five three million adults in Albania used the platform in a country of only two point four million people. This suggests that a large share of thirteen to eighteen-year-olds were active on TikTok as well. The platform's short video format, driven by its "For You" feed, viral challenges, and interactive filters, proved especially compelling to adolescents, who value quick peer feedback and ambient entertainment.

Yet TikTok's success also raised concerns about data privacy, mental health, and online harms. Every view, like, and comment contributes to a complex profiling engine designed to keep users engaged. Teenagers may not fully grasp how these algorithms shape what they see, or collect sensitive information, ranging from device identifiers to location data. In response to many concerned parents and educators and following the stabbing of a fourteen-year-old in November 2024 that local media linked to social media conflicts, Prime Minister Edi Rama proposed restricting TikTok nationwide. A government survey of sixty-five thousand parents reportedly showed ninety percent in favor of such a measure.

On 6 March 2025, the Council of Ministers adopted a formal decision to interrupt access to TikTok for twelve months, citing its potential to incite violence and negatively affect children's psycho-social development, as well as concerns about personal data protection. The decision entered into force on 7 March, and on 13 March, the National Authority for Cybersecurity issued a letter instructing Internet service providers to block TikTok's Internet protocol addresses, domain name system records, server name indications, and domains owned by ByteDance. By the end of that day, providers confirmed that access had been technically interrupted across Albanian territory.

However, critics, including media freedom and digital rights organizations warned that the measure lacked transparency, proportionality, and legal precision. SCiDEV (2025) emphasized that no concrete data was published to support the ban's necessity or its effectiveness in protecting minors. Albania became the only country in Europe to enforce a full national ban on TikTok, contrasting with other countries that opted for age-verification systems or platform-specific oversight. Despite these efforts, early reports suggest that many teenagers continue to access TikTok on their mobile devices by means of virtual private networks or proxy applications.

### I.3 Rationale and Research Gap

A broad shutdown of the platform cannot resolve deeper challenges of digital literacy, data privacy awareness, or the psychological drivers behind teenagers' engagement online. International standards including the General Data Protection Regulation of the European Union, which requires parental consent for processing children's data, and the United Nations Convention on the Rights of the Child, which affirms a child's right to privacy, offer useful guidelines but rely on local adaptation and enforcement. Albania's decision to block access without launching a nationwide program to teach digital safety has drawn criticism from media-freedom organizations, which warn that limiting access without building understanding may restrict freedom of expression and simply push usage into hidden channels.

Most existing research on adolescent data awareness focuses on Western Europe or North America, often in controlled laboratory settings or through online surveys that assume stable high-speed connections. It remains unclear whether those findings carry over to Albanian youth, who contend with different network reliability, device sharing, and cultural norms. Likewise, small pilot schemes in Albania such as media-literacy lessons in thirty schools have shown only modest improvements and have not been evaluated at scale.

### I.4 Research Question

To what extent are Albanian minors aware of the data privacy risks associated with TikTok, and how do they engage with the platform before and after its national ban?

### I.5 Thesis Objectives and Purpose

This thesis aims to explore the intersection between adolescent behavior and data privacy, using TikTok as a case study. The main objectives are:

1. To analyse how minors use TikTok and what motivates their engagement with the platform.
2. Assess teenagers' knowledge of TikTok's data-collection practices and examine how that knowledge relates to their choice of email when setting up an account.
3. To evaluate Albania's current legal and institutional efforts in protecting minors online.
4. To determine whether and how minors continue to access TikTok after the ban, using tools such as VPNs.
5. To propose recommendations for education, policy, and digital literacy initiatives tailored to youth.

### I.6 Thesis Hypothesis

This study hypothesizes that the 2024 TikTok ban in Albania, while meant to protect minors from harmful content and data risks, may not have achieved its desired effect. First, it is likely that a significant number of adolescents continue to use the platform by means such as virtual private networks or other technical workarounds. This suggests that simply blocking the app might not work as intended. Second, we believe that those who understand that TikTok gathers their personal information will take more careful steps to protect their privacy,for example, by creating an account with a made-up email address than those who are unaware of how the application collects data. Lastly, the ban may have underscored a more fundamental issue: a lack of structured education on data security and digital among young Albanians. Without clear education on how their personal data is collected, shared, and kept safe, any policy no matter how well-meaning, is unlikely to ensure that teenagers use social media responsibly over the long term.

# II.Literature Review

## II.1 Awareness and Usage Patterns on TikTok

Research consistently shows that young people misunderstand how much personal information they share and how it is used. The EU Teens and Privacy Survey (2018) found that while more than eighty-five percent of European adolescents could locate basic privacy settings, time and location, to build detailed profiles. The Study on Adolescent Data Perception (2015) reported that many teenagers in the United States confuse hiding a post from friends with preventing third-party data collection, revealing a fundamental gap in understanding.

Laboratory investigations of TikTok further highlight this mismatch. The TikTok Recommendation Dynamics Lab Report (2022) showed that within ten minutes of engaging with a single category of video, the “For You” feed narrows sharply, yet participants rarely recognise this shift or its implications. Complementary qualitative work in the Qualitative Interview Report on TikTok (2021) found that high-schoolers often describe the platform as purely “for fun,” unaware that every interaction feeds a data-collection loop.

These findings underscore two important points. First, platforms refine their engagement algorithms at a rapid pace. Second, even digitally native teenagers lack awareness of the underlying data flows.

However, those studies were conducted in environments with reliable high-speed connections and may not reflect the realities of emerging digital markets like Albania, where network speed, device sharing, and local usage habits differ significantly.

Recent evidence supports this concern: The Youth in Digital Albania Survey (2022) revealed that fewer than 20% of teenagers have ever modified app permissions or reviewed the privacy settings on platforms like TikTok.

Moreover, in Albania, regulatory decisions such as the 2025 TikTok ban have focused on limiting access rather than enhancing awareness. As SCiDEV (2025) argues, efforts to restrict platforms without addressing the root causes of digital misunderstanding risk overlooking the real problem: teenagers continue to use these platforms, often through circumvention tools, without better understanding the implications of their digital footprints.

Theoretical models provide further insight. Nissenbaum's concept of "contextual integrity" explains that users often judge privacy breaches based on how information flows align with their expectations which TikTok's opaque algorithm disrupts. Boyd (2014) adds that teens tend to define privacy in social terms, prioritizing peer perception over institutional surveillance. This social orientation leaves them particularly vulnerable to invisible, systemic data extraction.

**II.2 Local Context: Ban Implementation and Youth Perspectives**

The Social Media Usage and Privacy Awareness in Albanian Youth (2024) report presents focus-group findings from Durres and Tirana, showing that Albanian teenagers view privacy as simply hiding content from parents, not as safeguarding their personal information from corporations or government. Terms like metadata and cookies rarely appear in their vocabulary, and when they do, explanations remain superficial.

In the Analysis of TikTok Privacy Policy in Albanian (2021), researchers reviewed the official Albanian translation of TikTok's terms of service and found that more than eighty percent of the text requires a university-level reading ability far beyond the comprehension of most secondary-school students. Consequently, few teenagers consult or understand these documents, leaving them unable to enact the settings meant to protect them.

When the National Authority for Cybersecurity issued its blocking order on thirteen March 2025, technical details were scarce. Orkidea Xhaferraj of the Centre for Science and Innovation for Development noted that large-scale filtering of server name indications may require specialized infrastructure and extended implementation time, and that public information on these measures remains limited. Media-freedom alliances criticized the ban as excessive, arguing that it infringes on citizens' rights to information and freedom of expression.

The SCiDEV working paper further critiques the 2025 ban for its lack of proportionality and transparency, noting that the government provided no detailed data on the harm TikTok caused to minors or whether the ban would meaningfully reduce such harm. Although protection of children was cited as the main motive, the implementation relied on opaque technical measures like SNI filtering and did not clearly define what outcomes would trigger a re-evaluation of the decision.

This policy gap is particularly striking when compared to Albania's updated legal framework. In 2025, Law No. 25/2024 "On Cybersecurity" entered into force, mandating registration of critical digital infrastructure, creation of incident response teams (CSIRTs), and reporting of cyber incidents, including those involving minors. However, the law lacks clear mechanisms for integrating youth protection principles into platform-specific actions, leaving a disconnect between high-level cybersecurity governance and daily online experiences of adolescents.

Furthermore, Albania's approach stands apart from other European responses that emphasize platform accountability and parental consent mechanisms rather than blanket prohibitions. This discrepancy underscores a key weakness in current Albanian policy: the absence of a holistic digital protection strategy rooted in both legal clarity and youth education.

Despite the shutdown, surveys suggest that most teenagers continue to use TikTok via virtual private networks or proxy applications. This ongoing access highlights the need to study not only usage patterns but also whether awareness of data practices has changed.

**II.3 Regulatory Responses and Digital-Literacy Models**

International standards seek to protect minors online through both regulation and education. The European Union General Data Protection Regulation Children's Compliance Review (2016) mandates parental consent for processing data belonging to users under sixteen and requires privacy notices written in clear, age-appropriate language. The United Nations Convention on the Rights of the Child (1989) similarly enshrines the right to privacy in digital spaces. However, enforcement resources vary widely across countries.

Many European nations pair regulation with formal education. The United Kingdom Online Safety Impact Assessment (2021) funds the integration of data privacy, algorithmic fairness, and critical thinking into secondary-school curricula. Germany's Click safe Annual Program Report (2020) describes workshops where teenagers learn to adjust privacy settings, recognize phishing attempts, and understand their ongoing digital footprint. Estonia's Tiger Leap Initiative (2021) takes an even earlier approach, embedding basic coding and privacy lessons into primary-school teaching.

OECD (2023) research highlights that digital well-being education is most effective when introduced before adolescence. Programs in Finland and Denmark, such as Smart Kids, Safe Networks integrate gamified digital ethics into early curricula. Albania, by contrast, has delayed structural investments in this area.

In Albania, a UNESCO-supported pilot study introduced media-literacy modules in thirty schools in 2022. Evaluations reported a modest fifteen percent improvement in students' ability to interpret privacy notices, but the program has not been expanded nationwide. Civil society has stepped in with projects such as the Albanian Media Institute's "Learn to Discern" workshops, which have reached about two thousand students in Tirana and Shkodër. Nevertheless, these remain stand-alone efforts without integration into a national curriculum or a system for tracking long-term impact.

In 2025, Albania also adopted Law No. 124/2024 "On Personal Data Protection", harmonizing its legislation with the GDPR. The law introduced Data Protection Impact Assessments (DPIAs), 72-hour breach notifications, and stricter penalties for unlawful processing. Yet, implementation remains uneven, and the law does not mandate age-appropriate notices or child-specific compliance protocols beyond general parental consent.

SCiDEV (2025) recommends that Albania move toward a "Better Regulation" framework aligned with EU norms, combining transparent legal measures with systemic investment in media and information literacy (MIL) and cybersecurity education, starting in the pre-university system for both students and teachers.

**Gap and Implications:** Relying on one-off workshops and broad platform bans exposes a critical weakness in ensuring teenagers understand digital risks. Unless minors learn both why applications gather personal information and how to manage their digital presence responsibly, technical restrictions will offer only temporary relief and may even lead to riskier, hidden behaviors. The Albanian example demonstrates that in the absence of a rights-based and educationally grounded approach, restrictions may not only fail but also introduce new risks to democratic norms and children's online safety.

This thesis contributes by filling that gap, linking behavioral data from Albanian teenagers post-ban to the evolving regulatory landscape. It explores whether legislative action alone (e.g., cybersecurity laws and data protection mandates) can shift digital behavior, or whether integrated education and rights-based literacy remain essential.

# III.Methodology

This chapter outlines the research approach used to examine two interconnected phenomena: (1) TikTok's data-handling practices regarding minors in Albania, and (2) the behavioral, emotional, and perceptual responses of Albanian youth following the 2024 national ban. A mixed quantitative design was adopted, grounded in regulatory analysis, risk modeling, and cross-sectional survey methods.

## III.2 Research Design and Rationale

The study employed a dual-track quantitative design, combining policy analysis with survey-based behavioral measurement. This structure enabled a holistic assessment of both systemic and individual-level data practices and perceptions.

First, a data security assessment was conducted to analyze TikTok's privacy policies, before and after the 2024 Albanian ban, against national and international legal standards. The legal documents reviewed included Law No. 124/2024 on Personal Data Protection, Law No. 25/2024 on Cybersecurity, and Law No. 18/2017 on Children's Rights. European instruments such as the General Data Protection Regulation (GDPR) were also referenced. Risk analysis followed the **ISO/IEC 27005** and **NIST SP 800-30** frameworks, identifying vulnerabilities in location data, algorithmic manipulation, and behavioral tracking in the absence of effective consent mechanisms for minors.

Second, a cross-sectional survey was conducted among Albanian minors aged 12 to 18, using a structured questionnaire to evaluate TikTok usage patterns, privacy awareness, emotional response to the platform ban, and attitudes toward policy interventions. This approach allowed for the quantitative examination of correlations between digital behavior, emotional state, and awareness of privacy threats.

### III.3 Participants and Sampling

Survey data were collected from 203 students at two public educational institutions in Tirana:

- *Myslym Keta High School* (n = 159; ages 15–18)
- *Betim Muço Elementary School* (n = 44; ages 12–14)

A purposive sampling method was used to ensure representation across early and late adolescent groups. Prior to survey administration, schools were contacted and consent was obtained from administrators. Participation was voluntary and anonymous, with students informed of their right to withdraw at any time. No personal identifiers (names, IPs, device metadata) were collected.

### III.4 Variables and Survey Instrument

The survey instrument was designed in line with best practices for adolescent digital behavior studies and included both categorical and ordinal variables. Key dimensions captured were:

- Usage Patterns: Ownership of a TikTok account before the ban, daily usage time, and use of VPNs or proxies.
- Privacy Behavior: Type of email used to register (e.g., personal, fake, or parent-owned).
- Awareness: Self-reported knowledge of TikTok's data collection practices.
- Emotional Impact: A 5-point Likert scale measuring emotional state when TikTok was inaccessible (1 = drained, 5 = relaxed).
- Attitudinal Measures: Student opinions on the fairness and effectiveness of the ban, and whether they had ever reported content.

The survey was pre-tested with 12 students not included in the final sample to ensure clarity and age-appropriate phrasing. Minor revisions were made before final distribution.

### III.5 Data Collection and Analysis Tools

Data were collected in June 2025 using printed forms administered in supervised classroom sessions. Responses were digitized and analyzed using RStudio, with cleaning via the janitor package and further transformation using dplyr.

Three main statistical approaches were employed:

1. Descriptive statistics to report frequencies of continued usage, VPN access, and platform engagement post-ban.
2. Inferential testing to evaluate hypotheses:
    - A one-proportion z-test was used to test whether the majority of students continued accessing TikTok after the ban.

- A chi-square test of independence examined the association between privacy awareness and protective behavior (e.g., using fake emails or restricting app permissions).
- A one-way ANOVA tested differences in emotional responses across varying usage intensity groups.

Visualizations were created using ggplot2, including bar charts, histograms, boxplots, and forest plots for model interpretation. Reproducibility was ensured by using only open-source tools and documenting all scripts.

### III.6 Ethical Considerations

This study adheres to the ethical standards for research involving minors in educational settings. All participants were informed in advance about the nature of the study. No incentives were provided to prevent coercion. The survey was fully anonymous, and the data were stored in a secure, encrypted environment accessible only to the principal investigator.

Parental consent was not explicitly required due to the non-sensitive nature of the study and the anonymization of all data, in accordance with Article 8 of the GDPR and Law No. 18/2017 on Children's Rights in Albania.

### III.7 Limitations

While the survey covered two schools in Tirana, the sample may not fully represent the national demographic or students in rural areas. Additionally, emotional state and awareness were self-reported and may be affected by subjective biases. However, the findings provide a grounded starting point for understanding post-ban behavior in the Albanian context.

***Note:** This methodology chapter describes the exclusively **survey-based** approach used to analyze Albanian minors' behavior and perceptions regarding TikTok's ban and data practices. All subsequent chapters present the results and their implications.*

# IV.Behavior of Minors on TikTok

## IV.1Introduction

While the Albanian government framed its TikTok ban as a protective measure, the real impact on adolescent behavior and emotional well-being remains underexplored. Drawing on survey data from 203 students across two Tirana schools, this section investigates how minors continued accessing the platform, how they felt in its absence, and what factors sustain their engagement in the face of legal and technical barriers.

The analysis focuses on three core dimensions: (1) the persistence and intensity of TikTok usage despite the ban, including the use of VPNs or proxy applications; (2) the emotional responses associated with platform restriction, ranging from stress to indifference; and (3) the social and algorithmic drivers that continue to shape youth behavior in digital spaces. These findings contribute to a broader understanding of how Albanian minors navigate technological controls, and offer insight into the limitations of enforcement-focused regulatory models.

In doing so, the chapter provides empirical grounding to ongoing debates about digital platform governance, adolescent media habits, and the importance of integrating behavioral insights into policy design.

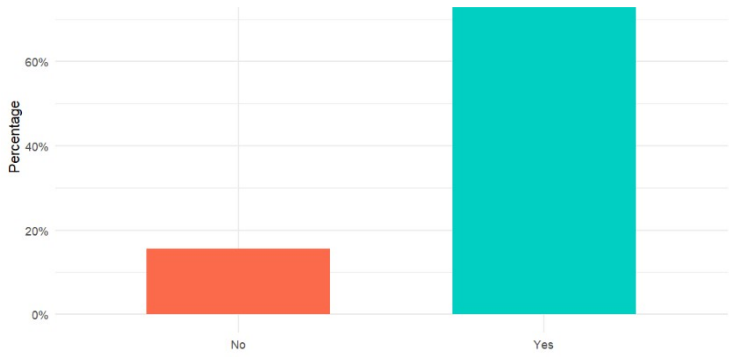
## IV.2Hypothesis-Driven Results

### *IV.2.1Hypothesis 1: Post-Ban TikTok Usage*

**$H_0$**: The proportion of Albanian minors who continue to use TikTok after the national ban is equal to or less than 50%.
**$H_a$**: The proportion of minors continuing to use TikTok exceeds 50%.

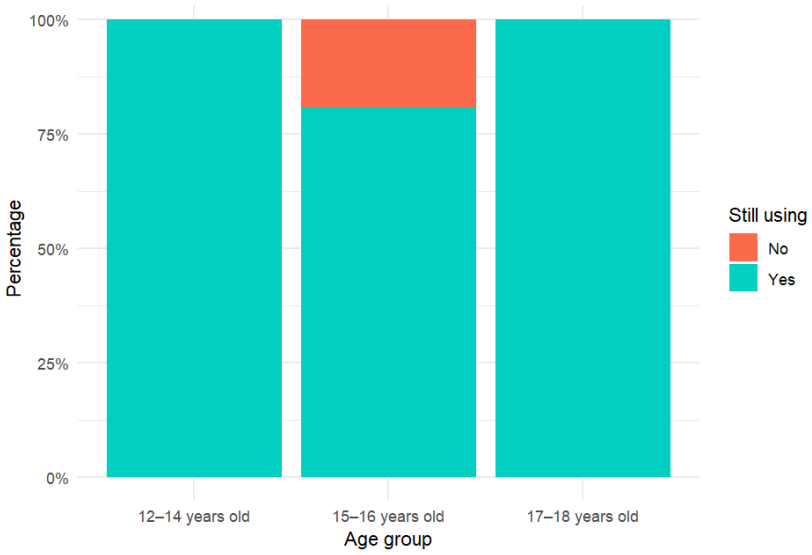
Survey responses reveal that **84.5%** of minors continue to access TikTok despite the national ban, primarily through **VPNs or alternative clients**.
A one-proportion test confirmed that this percentage is statistically greater than 50% ($\chi^2 = 90.28$, $p < 0.0001$), with a 95% confidence interval ranging from 79.4% to 100%.

*1.Figure 4-1.Overall percentage still using TikTok post-ban*




*Source:Author Findings,2025*

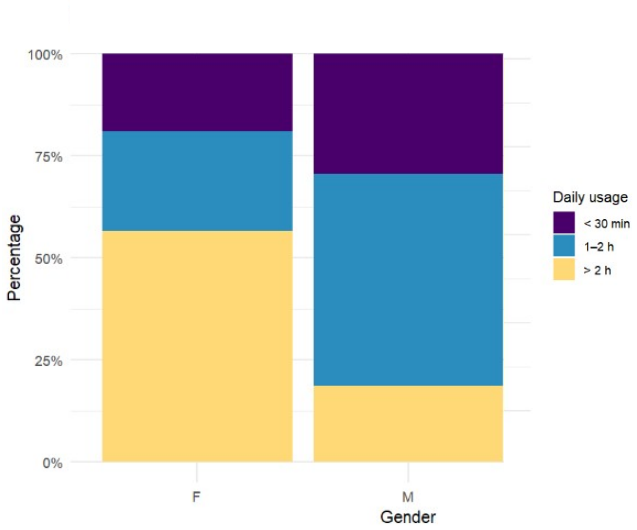
*2.Figure 4-2.Tiktok usage after ban by age group*




*Source:Author Findings,2025*

Further breakdowns show that **100%** of students aged 12–14 and 17–18 reported continued use, compared to **81%** in the 15–16 age group.

*3.Figure 4-3.Tiktok usage intensity by gender*




*Source:Author Findings,2025*

**of** Gender differences are also notable: **86.6% females** versus **74% of males** report ongoing usage.

### *IV.2.2Hypothesis 2: Privacy Awareness and Protective Behavior*

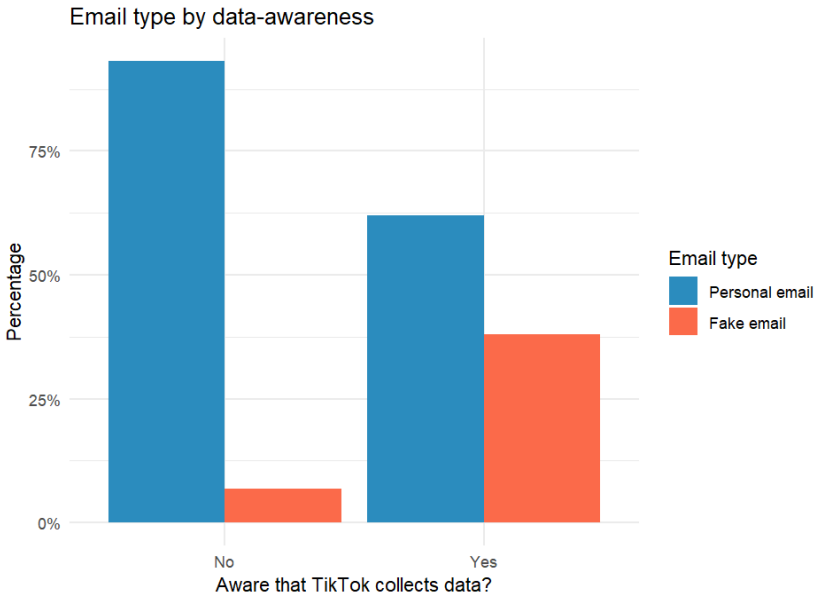
The majority of students reported using TikTok for 1–2 hours per day, with over 25% exceeding two hours. Accounts.

**$H_0$**: Awareness of TikTok's data collection practices is not associated with increased use of fake email accounts.

**$H_a$**: Students who are aware that TikTok collects data are more likely to use fake email addresses.

Among respondents who acknowledged that TikTok collects personal data, a larger proportion reported using **fake or secondary email addresses** during account registration. A chi-square test of independence yielded $\chi^2 = 8.06$, $p = 0.0045$, indicating a statistically significant association.

*4.Figure 4-4.Email type by data-awareness*




*Source:Author Findings,2025*

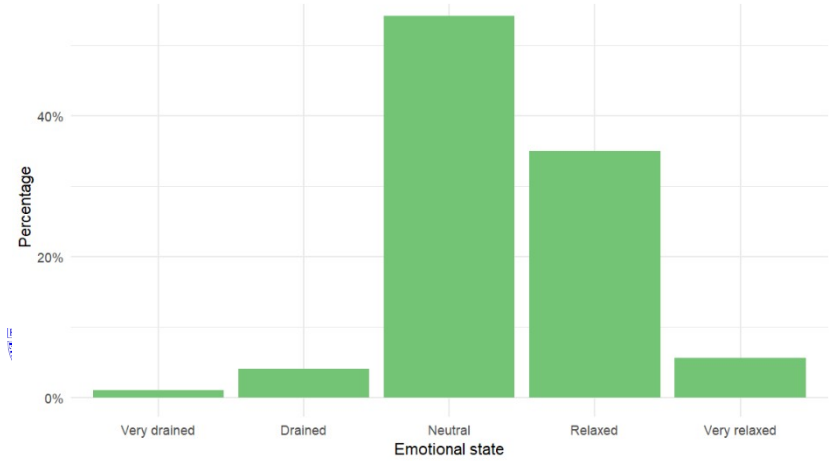
**Conclusion**: The null hypothesis is rejected. Awareness appears to foster **privacy-protective behavior**, especially among older students. This finding supports the idea that digital literacy can translate into concrete privacy actions among minors.

### *IV.2.3Hypothesis 3: Emotional Impact and Usage Intensity*

$H_0$: There is no relationship between TikTok usage intensity and self-reported emotional state.
$H_a$: TikTok usage intensity is associated with variation in emotional states.

To evaluate emotional impact, students rated how they typically feel after using TikTok on a scale from 1 ("Very drained") to 5 ("Very relaxed"). The distribution was nearly even: 30% reported feeling relaxed, 40% neutral, and 30% drained or anxious.

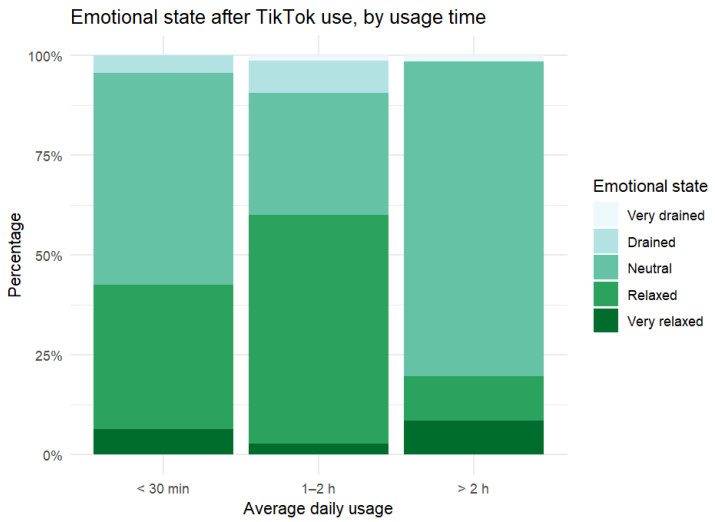
*5.Figure 4-5.The emotional state after using Tik Tok*



*Source:Author Findings,2025*

A one-way ANOVA tested whether emotional states varied significantly by usage intensity. The result ($F(2, 190) = 2.74$, $p = 0.0672$) was not statistically significant at the 0.05 threshold, but it approached significance, suggesting a trend worth further investigation.

*6.Figure 6.6 The emotional state by usage time*




*Source:Author Findings,2025*

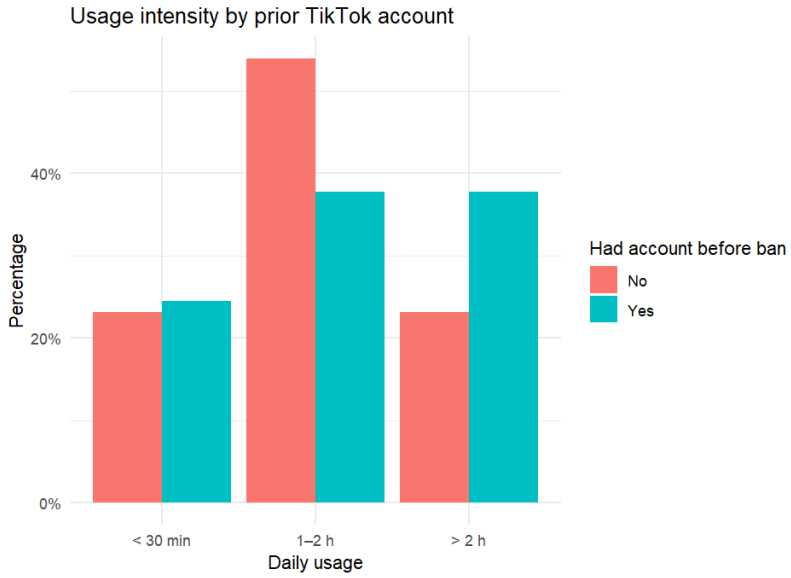
**Conclusion**: The null hypothesis is not rejected. However, the variability among heavy users points to possible emotional volatility linked to extended platform use. A larger or longitudinal sample may yield stronger conclusions.

## IV.3 Additional Findings

### *IV.3.1 Daily Usage Patterns*

Most respondents use TikTok for 1–2 hours daily, while over 25% exceed two hours. A chi-square test revealed no significant link between prior account ownership and current intensity ($\chi^2(2) = 2.27$, $p = 0.322$), suggesting new and long-time users display similar behavior.

*7.Figure 4.7 Usage intensity by prior TikTok account*




*Source:Author Findings,2025*

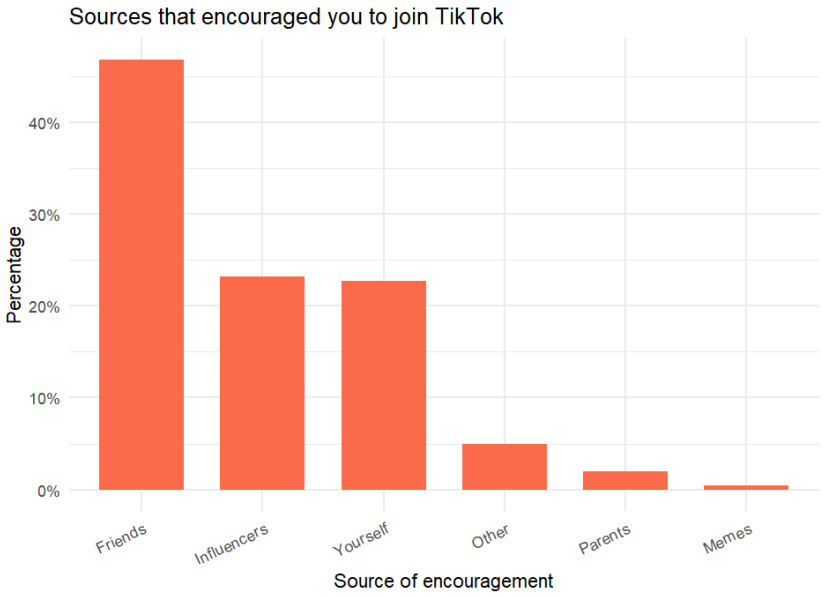
### *IV.3.2 Age of First Adoption of TikTok*

The majority of students created their first TikTok account between the ages of 13 and 15, a period considered critical for media exposure and identity development. This timing suggests that early adolescence is a strategic window for implementing digital literacy education and family-level controls.

### *IV.3.3 Peer and Influencer Influence*

When asked who encouraged them to join TikTok, 50% of students cited peers, 25% influencers, and the remainder parents or others. These findings reflect diffusion-of-innovation patterns, where social proof and trend adoption drive platform uptake.

*8.Figure 4-8.Bar chart of "Who encouraged you to join TikTok?"*



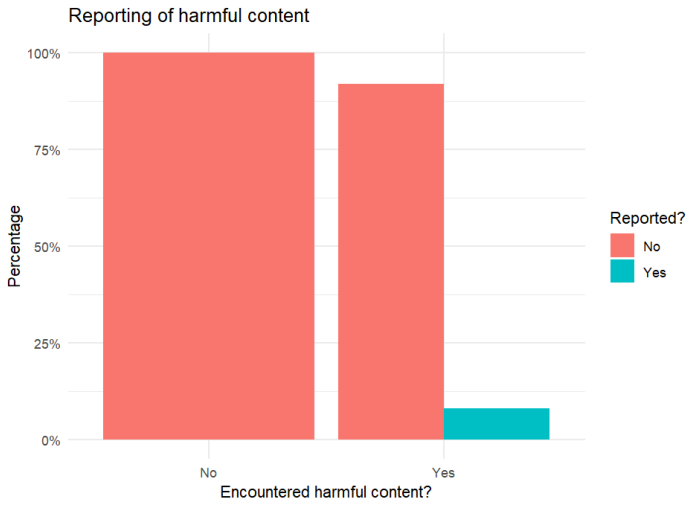

*Source:Author Findings,2025*

### *IV.3.4 Exposure vs. Reporting of Harmful Content*

Although **60%** of respondents said they had seen harmful or inappropriate content, only **5%** of these individuals had ever reported it.

A Fisher's exact test yielded $p < 0.001$, indicating a significant **gap between awareness and action**.

*9.Figure 4-9. Grouped bar of reporting behavior among those exposed to harmful content*



This discrepancy suggests that while adolescents may recognize violations, they rarely engage with platform safety tools, potentially due to distrust, apathy, or poor interface design.

### IV.4 Summary of Findings

The results indicate that 84.5% of minors continue to use TikTok despite the national ban, primarily through VPNs or alternative tools. Peer encouragement was the most common factor influencing adoption. Emotional responses to TikTok varied by usage intensity, with some students reporting fatigue and others calmness. A statistically suggestive ($p = 0.0672$) relationship between time spent on the app and emotional state was observed. Additionally, awareness of TikTok's data collection correlated with increased use of fake emails, suggesting limited adoption of privacy-preserving behavior. Most students who encountered harmful content did not report it. Overall, behavioral adaptation, rather than disengagement. characterized minors' responses to the ban.

## V.DATA SECURITY IN SOCIAL MEDIA

### V.1TikTok's Privacy Architecture: Policy and Risk Landscape

TikTok, developed by ByteDance Ltd., presents a robust data extraction and behavioral profiling architecture, often masked under the interface of short-form entertainment. According to its global privacy policy, the platform collects a vast array of data: device identifiers, geolocation, usage patterns, browsing history, and even biometric markers such as faceprints and voiceprints. While the policy claims that consent is required for biometric collection in protected jurisdictions, critics argue that these consent mechanisms are vague and often misunderstood by users, especially minors.

Minors frequently lack the legal and cognitive capacity to provide meaningful consent, a requirement under the EU General Data Protection Regulation (GDPR).While TikTok nominally restricts access to users under 13, enforcement is inconsistent, and age-verification protocols are easy to bypass. As such, personal data is routinely collected from underage users, often without parental approval or adequate notice.

The volume and type of personal information collected by TikTok can present a wide range of risks, particularly when the primary users are children and adolescents. While adults may better understand the digital environment and take steps to protect themselves, minors often lack the tools, awareness, or maturity to assess how their information may be used, stored, or even exposed.

Several data protection frameworks, including ISO/IEC 27005 and NIST SP 800-30, outline how to assess risks based on the nature of the data, the threats it may face, and the vulnerabilities in the systems that handle it. When applying this type of assessment to TikTok's use among minors, a number of high-risk issues come into focus.

First, the volume of identifiable information, including location, device activity, and even biometric traits, creates a rich profile that can be exploited if accessed by unauthorized actors. This includes cybercriminals who may target underage users or use personal details for scams, phishing attacks, or identity theft. While TikTok stores this information to personalize the app experience, its centralization becomes a vulnerability in the event of a breach.

Second, the lack of meaningful consent from minors is a major problem. Children under 16 often agree to terms without understanding them, and age verification mechanisms remain weak in many cases. This means that even though legal frameworks like the GDPR require verifiable parental consent for data collection from minors, enforcement on TikTok remains inconsistent. As a result, personal data from underage users is often collected in ways that do not fully comply with regional or international standards.

Third, the behavioral profiling done through algorithms carries long-term implications. TikTok's system tracks what users watch, how long they pause, what they interact with, and what they skip. For minors, this not only creates filter bubbles that limit the content they see but also allows the platform to build a psychological profile over time. This level of profiling may be used to influence user behavior in ways that are subtle and manipulative, raising ethical concerns, especially when the users are too young to recognize that they're being targeted.

Moreover, third-party access is another key concern. Like many tech platforms, TikTok shares some user data with advertisers and service partners. Even when anonymized, aggregated data can sometimes be de-anonymized, especially when combined with other datasets. For minors, who may reveal personal details more openly or link accounts with other apps, this risk is magnified.

Finally, data retention policies raise questions about how long this information is kept and for what purpose. If TikTok continues to store minor user data even after a user deletes their account or stops using the platform, the exposure window remains open far longer than it should.

Taken together, these concerns reveal a deeply integrated data ecosystem that monetizes attention and behavior at the expense of transparency and child protection. This underscores the urgent need for localized digital literacy programs, age-specific safeguards, and enforcement mechanisms in Albania that move beyond blanket bans.

In combination, these factors expose minors to both immediate risks like stalking, online exploitation, or doxxing and long-term harms, such as psychological profiling, loss of privacy, or misuse of their data in adulthood. While some countries have introduced bans or strengthened regulation in response to these concerns, the Albanian ban in 2024 illustrates how reactive policy measures alone cannot fully address these deeply embedded risks.

What emerges is the need for proactive protection: not just technical restrictions or temporary bans, but clear, transparent data-handling practices and stronger education for both youth and parents about what digital privacy actually means in today's algorithm-driven environments.

### V.2 Institutional Responses and Enforcement Gaps

Across Europe, regulatory investigations and legal enforcement have increasingly exposed structural weaknesses in TikTok's approach to safeguarding minors. One of the most critical failures involves the platform's deficient age-verification systems. In 2023, the UK's Information Commissioner's Office (ICO) fined TikTok £12.7 million for permitting approximately 1.4 million children under the age of 13 to access the platform without verifiable parental consent, a clear violation of both national standards and the EU General Data Protection Regulation (ICO, 2023). Compounding the issue, newly registered teenage accounts were by default set to "public," thereby facilitating exposure to unwanted contact and elevating the risk of exploitation.

In parallel, the platform's algorithmic infrastructure has raised further concerns. TikTok's recommendation engine, designed to maximize user engagement, has repeatedly been found to promote emotionally charged or harmful content, including videos related to self-harm and dangerous viral challenges. In 2024, Italy's data protection authority issued a €10 million fine after concluding that such content had been served to minor users in the absence of sufficient content moderation or protective filtering (Garante per la Protezione dei Dati Personali, 2024).

These findings reinforce critiques that TikTok's algorithmic logic privileges engagement over well-being, particularly for vulnerable user groups.

Moreover, the issue of unlawful cross-border data transfers has deepened regulatory scrutiny. In 2025, Ireland's Data Protection Commission (DPC) imposed a record €530 million fine on TikTok for enabling European user data, including that of minors, to be accessed from servers located in China, contravening GDPR safeguards on international data flows (DPC, 2025). This came shortly after a separate €345 million penalty in 2023 for failing to process children's data lawfully. Despite TikTok's stated commitments to regional data hosting and compliance, these cases suggest systemic lapses in both transparency and accountability.

These institutional responses reflect a broader shift from reactive oversight to proactive enforcement within the EU. Regulatory bodies are increasingly mandating algorithmic audits, age-appropriate design requirements, and higher standards for consent verification. However, such frameworks remain underdeveloped in the Albanian context. Although Albania implemented a national ban on TikTok in 2024, this measure has lacked the infrastructural and legal support necessary to ensure its effectiveness. Without independent auditing capabilities, improved data governance, or digital literacy education, the ban has done little to mitigate the underlying risks. Instead, it highlights the limitations of symbolic regulation in the face of complex, cross-border digital ecosystems.

# VI.Conclusions and Recommendations

This thesis set out to explore how Albanian minors navigate data privacy risks on TikTok, particularly in the aftermath of the national platform ban. Drawing on a combination of regulatory analysis and empirical survey data, the findings challenge the assumption that prohibition alone is a sufficient strategy to safeguard adolescents in digital spaces.

Despite the March 2025 decision by Albanian authorities to block TikTok on grounds of child protection and national cybersecurity, the platform remains widely used. An overwhelming 84.5% of minors surveyed reported continuing to access TikTok through VPNs or alternate apps. This suggests that bans devoid of enforcement mechanisms or digital literacy efforts function more as political symbols than effective tools of protection. Instead of disengaging, teenagers adapt, bypassing technical blocks while maintaining or even intensifying their engagement.

The behavioral data also reveals significant variation in emotional response to TikTok usage. While some adolescents reported feeling calm or entertained, others expressed fatigue or emotional drainage. Although the association between usage intensity and emotional impact did not reach statistical significance, the trend points to an important nuance: adolescents are not passive consumers, but emotionally affected participants in an algorithmic environment they do not fully understand. The role of algorithmic amplification, particularly in promoting emotionally intense or risky content, remains a persistent area of concern, especially in light of documented enforcement gaps at the international level.

Moreover, the data indicates that privacy awareness, while limited, is not entirely absent. Minors who demonstrated awareness of TikTok's data collection practices were significantly more likely to use fake or secondary emails, reflecting selective protective behavior. Yet, paradoxically, the same minors were unlikely to report harmful content. This points to a troubling "knowledge-action" gap, where adolescents recognize risks but fail to act due to mistrust in reporting mechanisms, low expectations of outcomes, or simple resignation. Protection, in this context, is improvised, fragmented, and often insufficient.

Institutionally, Albania has made key legal updates, including the 2025 laws on cybersecurity and personal data protection. However, these frameworks remain decoupled from the daily realities of adolescent digital life. Regulatory action has focused heavily on infrastructure and institutional reporting, while overlooking the cognitive, emotional, and behavioral layers of youth engagement.

Unlike countries that pair enforcement with early digital education, Albania's strategy has leaned toward restriction without empowerment.**Therefore, this thesis recommends a reorientation of child data protection policy from prohibition to participation.**

First, digital literacy must be integrated into the national curriculum at all levels. Adolescents need structured, rights-based education that helps them understand how platforms function, how data is collected, and how consent can be meaningfully exercised. Second, enforcement mechanisms must be accompanied by transparency and clear criteria. If bans are used, they should be time-bound, independently reviewed, and paired with parallel investments in civic education. Third, Albania must move toward regulatory harmonization with the EU's evolving standards especially in areas like algorithmic transparency, age-appropriate design, and cross-border data transfers. Without this alignment, minors will remain exposed to invisible risks in a globally networked environment.

Albanian minors are not unaware or indifferent. They are improvising privacy on platforms designed to keep them engaged and exposed. The role of the state should not be to block access in isolation, but to ensure that when adolescents go online, they do so with informed agency, institutional protection, and a clear understanding of their digital rights.

# VII. Future Outlook

As Albania and other countries face the challenges of growing up in a digital world, the conversation must move beyond quick bans and toward thoughtful, practical approaches. The findings of this thesis show that teenage digital habits are not fixed and cannot be easily changed by simply blocking apps. Instead, they are shaped by peer groups, algorithmic systems, and their growing, but still limited, understanding of how the online world truly functions.

Future research should investigate how patterns develop over time. A one-time survey offers a useful snapshot, but it cannot illustrate how habits, emotions, and awareness of risk evolve. Longitudinal studies could determine whether awareness campaigns have a lasting effect or whether students eventually grow indifferent to them. Moreover, more personal methods, such as interviews, digital diaries, or participatory research, could help uncover the reasoning behind teenage choices: why some opt for fake emails, why they avoid reporting tools, and how they feel emotionally and socially about restrictions.

From a policy perspective, Albania should not remain isolated in its approach. As the European Union continues to implement the Digital Services Act (DSA) and frameworks for age-appropriate platform design, Albania should align itself with these evolving standards. Cooperation with regional partners can facilitate knowledge sharing, enhance enforcement capacity, and promote a more consistent response to cross-border digital threats.

Most importantly, digital governance must be intergenerational. Rather than casting adults solely as protectors and adolescents as passive recipients of protection, future frameworks should recognize young people as collaborators in building a safer digital environment. This includes involving them in national policy-making processes, evaluating tools through their perspective, and treating digital literacy not merely as instruction, but as active civic engagement.

Ultimately, the future of data privacy for minors in Albania will not be secured through firewalls or content filters alone. What is needed is a shift in mindset: from attempting to control what young people can access, to preparing them for what they will inevitably encounter. The adolescent users of tomorrow require not only protection from data misuse, but also preparation for digital citizenship. This thesis offers a first step toward achieving that goal.

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**APPENDIX**

**Appendix A:Code in R**

```
# 0. Install needed packages (run once)
install.packages(c("readxl","dplyr","janitor","ggplot2","forcats","MASS","scales","broom"))

# 1. Load required libraries
library(readxl)
library(dplyr)
library(janitor)
library(ggplot2)
library(forcats)
library(MASS)
library(scales)
library(broom)
library(stringr)

# 2. Read and clean Excel data
df <- read_excel("C:/Users/selam/OneDrive/Desktop/survey/survey.xlsx", sheet = 1) %>%
  clean_names()

# 3. Recode variables
df <- df %>%
  mutate(
    account_before =
factor(a_ke_pasur_nje_llogari_personale_ne_tik_tok_para_ndalimit_te_tij_ne_shqiperi, levels =
c("Jo", "Po")),
    still_using =
factor(nese_po_a_vazhdon_ta_perdoresh_tik_tok_permes_vpn_ose_aplikacioneve_te_tjera,
levels = c("Jo", "Po")),
    usage_time = factor(sa_kohe_kalon_mesatarisht_duke_perdorur_tik_tok, levels = 1:3, labels =
c("< 30 min", "1–2 h", "> 2 h"), ordered = TRUE),
    first_age = sa_vjec_ishe_kur_krijove_llogarine_tende_ne_tik_tok,
    aware_data =
factor(a_e_di_qe_tik_tok_mbledh_te_dhenat_e_tua_personale_si_numrin_e_telefonit_emailin_v
endndodhjen, levels = c("Jo", "Po")),
    email_type = factor(cilin_email_ke_perdorur_per_te_krijuar_llogarine_ne_tik_tok, levels =
c("Email personal", "Email i prindërve", "Email i rremë (fake)")),
    fake_email = ifelse(cilin_email_ke_perdorur_per_te_krijuar_llogarine_ne_tik_tok == "Email i
rremë (fake)", 1, 0),
    saw_bad =
factor(a_ke_pare_ndonje_permbajtje_ne_tik_tok_qe_mendon_se_jep_shembuj_te_gabuar_apo_t
e_rrezikshem_per_moshen_tende, levels = c("Jo", "Po")),
    reported_bad =
factor(nese_po_a_ke_raportuar_ndonjehere_nje_permbajtje_te_papershtatshme, levels = c("Jo",
"Po")),
```

```
    calm_num = si_ndihesh_zakonisht_pas_perdorimit_te_tik_tok,
    support_res =
factor(mendon_se_duhet_te_kete_me_shume_kufizime_per_perdorimin_e_tik_tok_nga_femijet_
dhe_adoleshentet, levels = c("Jo", "Nuk i jam i/e sigurt", "Po")),
    fairness =
factor(sa_i_drejte_mendon_se_ishte_vendimi_per_te_ndaluar_tik_tok_un_ne_shqiperi,


levels = 1:5, labels = c("Shumë i padrejtë", "I padrejtë", "Neutral/e", "I drejtë", "Shumë i
drejtë"), ordered = TRUE),
    should_return = factor(a_mendon_se_tik_tok_duhet_te_rikthehet_ne_shqiperi, levels = 1:5,
labels = c("Absolutisht jo", "Jo", "Neutral/e", "Po", "Absolutisht po"), ordered = TRUE),
    age_group = factor(sa_vjec_je, levels = c("12–14 vjeç", "15–16 vjeç", "17–18 vjeç")),
    gender = factor(gjinia, levels = c("F", "M", "Tjetër")),
    vpn_num =
ifelse(tolower(as.character(nese_po_a_vazhdon_ta_perdoresh_tik_tok_permes_vpn_ose_aplikaci
oneve_te_tjera)) == "po", 1,

ifelse(tolower(as.character(nese_po_a_vazhdon_ta_perdoresh_tik_tok_permes_vpn_ose_aplikaci
oneve_te_tjera)) == "jo", 0, NA))
  )

# 4. Variable translations (Albanian to English)
df <- df %>%
  mutate(
    still_using_eng = recode(still_using, "Po" = "Yes", "Jo" = "No"),
    saw_bad_eng = recode(saw_bad, "Po" = "Yes", "Jo" = "No"),
    reported_bad_eng = recode(reported_bad, "Po" = "Yes", "Jo" = "No"),
    gender_eng = recode(gender, "F" = "Female", "M" = "Male", "Tjetër" = "Other"),
    email_type_eng = recode(email_type, "Email personal" = "Personal email", "Email i
prindërve" = "Parent's email", "Email i rremë (fake)" = "Fake email"),
    aware_data_eng = recode(aware_data, "Po" = "Yes", "Jo" = "No"),
    account_before_eng = recode(account_before, "Po" = "Yes", "Jo" = "No"),
    age_group_eng = recode(age_group, "12–14 vjeç" = "12–14 years old", "15–16 vjeç" = "15–
16 years old", "17–18 vjeç" = "17–18 years old")
  )

# 5. Hypothesis Testing
## H1: VPN use post-ban
prop.test(sum(df$vpn_num, na.rm = TRUE), sum(!is.na(df$vpn_num)), p = 0.5, alternative =
"greater")

## H2: Awareness vs fake email
email_test_df <- df %>% filter(!is.na(aware_data), !is.na(fake_email))
chisq.test(table(email_test_df$aware_data, email_test_df$fake_email))
```

```
## H3: Emotional state by usage intensity
df_clean <- df %>% filter(!is.na(calm_num), !is.na(usage_time))
emotion_model <- aov(calm_num ~ usage_time, data = df_clean)
summary(emotion_model)

# 6. Visualizations (Selected Outputs)
ggplot(df %>% filter(!is.na(still_using_eng)) %>% count(still_using_eng) %>% mutate(pct = n / sum(n)),
       aes(x = still_using_eng, y = pct, fill = still_using_eng)) +
  geom_col(width = 0.6) +



scale_y_continuous(labels = percent_format(accuracy = 1)) +
  labs(title = "Still Using TikTok After Ban", y = "Percentage", x = NULL) +
  scale_fill_manual(values = c("Yes" = "#00CFC1", "No" = "#FB6A4A")) +
  theme_minimal() +
  theme(legend.position = "none")
```

## Appendix B

### Appendix B.1-Original Survey(Albanian)

**Pyetësori për përdorimin e TikTok nga të miturit**

Pjesa 1: Informacioni bazë

1.Sa vjeç je?

☐ 12–14 vjeç
☐15–16 vjeç
☐17–18 vjeç

2.Gjinia:

☐ F
☐M
☐Tjetër

Pjesa 2: Përdorimi i TikTok

3. A ke pasur një llogari personale në TikTok para ndalimit të tij në Shqipëri?

☐ Po
☐Jo

4.Nëse po, a vazhdon ta përdorësh TikTok përmes VPN ose aplikacioneve të tjera?

☐ Po
☐Jo

5.Sa kohë kalon mesatarisht duke përdorur TikTok?

☐Më pak se 30 minuta
☐1–2 orë
☐Më shumë se 2 orë

6.Sa vjeç ishe kur krijove llogarinë tënde në TikTok?

☐ Më pak se 13 vjeç

☐13–15 vjeç
☐ Më shumë se 15 vjeç

7.Kush të ka nxitur të përdorësh TikTok?
☐ Shoqëria
☐ Prindërit
☐ Persona të famshëm/Influencer
☐ Tjetër: _____

Pjesa 3: Ndërgjegjësimi për sigurinë dhe privatësinë

8. A e di që TikTok mbledh të dhënat e tua personale (si numrin e telefonit, emailin, vendndodhjen)?
☐ Po
☐Jo

9.Cilin email ke përdorur për të krijuar llogarinë në TikTok?
☐ Email personal
☐ Email i prindërve
☐ Email i rremë (fake)

Pjesa 4: Mendimi personal dhe qëllimi i përdorimit

10. Për çfarë qëllimesh e përdor TikTok më së shumti?
☐ Argëtim
☐ Për të marrë informacione
☐ Për të njohur njerëz të rinj

11.A ke parë ndonjë përmbajtje në TikTok që mendon se jep shembuj të gabuar apo të rrezikshëm për moshën tënde?
☐ Po
☐Jo
☐Nuk i jam i sigurt

12.Nëse po, a ke raportuar ndonjëherë një përmbajtje të papërshtatshme?
☐ Po
☐Jo
☐Nuk i jam i sigurt

13.Si ndihesh zakonisht pas përdorimit të TikTok?
☐ Shumë i lodhur
☐ I lodhur
☐Neutral
☐ I qetë
☐Shumë i qetë

14.Mendon se duhet të ketë më shumë kufizime për përdorimin e TikTok nga fëmijët dhe adoleshentët?
☐ Po
☐Jo
☐Nuk i jam i sigurt

15. Sa i drejtë mendon se ishte vendimi për të ndaluar TikTok-un në Shqipëri?

☐ Shumë i padrejtë
☐ I padrejtë
☐Neutrale
☐I drejtë
☐Shumë i drejtë

16.A mendon se TikTok duhet të rikthehet në Shqipëri?

☐Absolutisht jo
☐Jo
☐Neutrale
☐Po
☐Absolutisht po

***Appendix B.2- Survey(English)***

**Title: TikTok Usage Questionnaire for Minors**

**Part 1: Basic Information**

1. How old are you?
☐ 12–14 years
☐ 15–16 years
☐ 17–18 years
2. Gender:
☐ Female
☐ Male
☐ Other

**Part 2: TikTok Usage**

3. Did you have a personal TikTok account before the platform was banned in Albania?
☐ Yes
☐ No
4. If yes, do you still use TikTok through VPNs or alternative applications?
☐ Yes
☐ No
5. On average, how much time do you spend on TikTok each day?
☐ Less than 30 minutes
☐ 1–2 hours
☐ More than 2 hours
6. How old were you when you created your TikTok account?
☐ Under 13
☐ 13–15
☐ Over 15
7. Who encouraged you to start using TikTok?
☐ Friends
☐ Parents
☐ Famous people/influencers
☐ Other: _____

**Part 3: Awareness of Safety and Privacy**

8. Are you aware that TikTok collects your personal data (e.g., phone number, email, location)?
☐ Yes
☐ No
9. What kind of email did you use to create your TikTok account?

☐ Personal email
☐ Parent's email
☐ Fake email

**Part 4: Personal Opinion and Platform Use**

10. What is your main reason for using TikTok?
☐ Entertainment
☐ Getting information
☐ Meeting new people

11. Have you ever seen content on TikTok that you believe is inappropriate or dangerous for your age?
☐ Yes
☐ No
☐ I'm not sure

12. If yes, have you ever reported inappropriate content?
☐ Yes
☐ No
☐ I'm not sure

13. How do you usually feel after using TikTok?
☐ Very drained
☐ Drained
☐ Neutral
☐ Calm
☐ Very calm

14. Do you think there should be more restrictions on TikTok use by children and teenagers?
☐ Yes
☐ No
☐ I'm not sure

15. How fair do you think the decision to ban TikTok in Albania was?
☐ Very unfair
☐ Unfair
☐ Neutral
☐ Fair
☐ Very fair

16. Do you think TikTok should be allowed again in Albania?
☐ Absolutely not
☐ No
☐ Neutral
☐ Yes
☐ Absolutely yes

## Appendix C: Statistical Output Tables and Charts

### *C.1 Hypothesis Testing Outputs*

*10.Table C.1: Proportion Test – Continued Usage of TikTok Post-Ban*

```
        1-sample proportions test with continuity correction

data:  sum(df$vpn_num, na.rm = TRUE) out of sum(!is.na(df$vpn_num)), null probability 0.5
X-squared = 90.28, df = 1, p-value < 2.2e-16
alternative hypothesis: true p is greater than 0.5
95 percent confidence interval:
 0.7940927 1.0000000
sample estimates:
        p
0.8445596
```

Test: One-sample proportion test
Null Hypothesis: 50% or fewer minors use TikTok post-ban
Result:
$X^2 = 90.28$, df = 1, p-value < 2.2e-16
Confidence Interval: [0.794, 1.000]
Estimated Proportion: 84.5%

*11.Table C.2: Chi-square Test – Data Awareness and Fake Email Usage*

```
> # H2: Privacy Awareness vs Fake Email
> email_test_df <- df %>% filter(!is.na(aware_data), !is.na(fake_email))
> chisq.test(table(email_test_df$aware_data, email_test_df$fake_email))

        Pearson's Chi-squared test with Yates' continuity correction

data:  table(email_test_df$aware_data, email_test_df$fake_email)
X-squared = 8.0587, df = 1, p-value = 0.004528

>
```

Test: Pearson's Chi-squared test
Result:
$X^2 = 8.06$, df = 1, p-value = 0.0045
Interpretation: There is a significant association between data awareness and use of fake emails.

*12.Table C.3: ANOVA – Usage Intensity vs. Emotional Impact*

```
>
> # H3: Usage Intensity vs Emotional Impact
> df_clean <- df %>% filter(!is.na(calm_num), !is.na(usage_time))
> emotion_model <- aov(calm_num ~ usage_time, data = df_clean)
> summary(emotion_model)
             Df Sum Sq Mean Sq F value Pr(>F)
usage_time    2   2.70  1.3515   2.738 0.0672 .
Residuals   190  93.77  0.4935
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
>
```

Test: One-way ANOVA
$F(2, 190) = 2.738$, p = 0.0672
Interpretation: Not statistically significant, but indicates a trend toward variation in emotional state by usage duration.

*13.Table C.4: Contingency Table – Usage Time vs Emotional State*

```
> table(df$usage_time, df$calm_num)

            1  2  3  4  5
  < 30 min  0  2 25 17  3
  1–2 h     1  6 23 43  2
  > 2 h     1  0 56  8  6
>
```

| Usage Time | 1 (Drained) | 2 | 3 (Neutral) | 4 | 5 (Relaxed) |
|---|---|---|---|---|---|
| **< 30 min** | **0** | **2** | **25** | **17** | **3** |
| **1–2 h** | **1** | **6** | **23** | **43** | **2** |
| **> 2 h** | **1** | **0** | **56** | **8** | **6** |